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(1) Arithmetic Operators

Arithmetic operators perform simple arithmetic operations like addition, subtraction, multiplication, division etc.,

Operator	Operation	Example
+	Addition	10 + 5 = 15
-	Subtraction	10 – 5 = 5
*	Multiplication	10 * 5 = 50
/	Division	10 / 5 = 2 (Quotient of the division)
%	Modulus (To find the reminder of a division)	10 % 3 = 1(Remainder of the division)

- The above mentioned arithmetic operators are binary operators which requires minimum of two operands.

Increment and Decrement Operators

++ (Plus, Plus) Increment operator

-- (Minus, Minus) Decrement operator

An increment or decrement operator acts upon a single operand and returns a new value. Thus, these operators are unary operators. The increment operator adds 1 to its operand and the decrement operator subtracts 1 from its operand. For example,

- **x++** is the same as **x = x+1;**
  It adds 1 to the present value of x

- **x--** is the same as to **x = x–1;**
  It subtracts 1 from the present value of x

The ++ or -- operators can be placed either as prefix (before) or as postfix (after) to a variable. With the prefix version, C++ performs the increment / decrement before using the operand.

For example: **N1=10, N2=20;**

S = ++N1 + ++N2;

The following Figure explains the working process of the above statement.

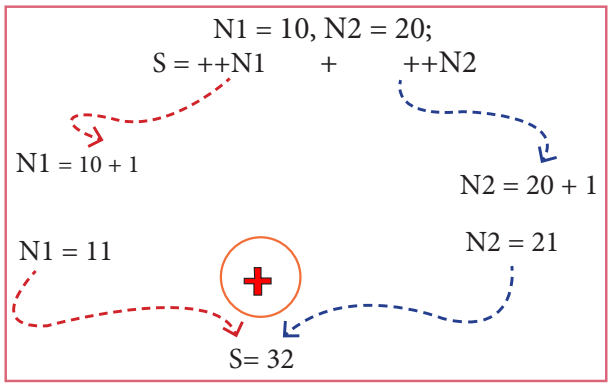

Figure 9.2 Prefix Increment Working process

In the above example, the value of num is first incremented by 1, then the incremented value is assigned to the respective operand.

With the postfix version, C++ uses the value of the operand in evaluating the expression before incrementing / decrementing its present value.

For example: **N1=10, N2=20;**

S = N1++ + ++N2;

The following Figure explains the working process of the above statement.